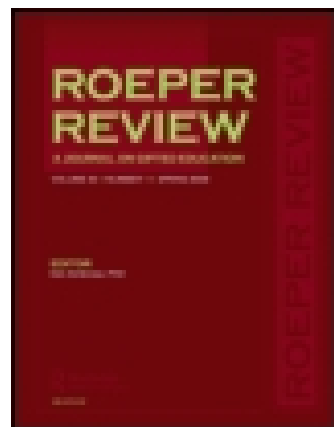


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Training analogical reasoning skills in the gifted

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definition clearly in mind we ask the children to explain why it is confusing when one person asks another *Should we have freedom of speech?* and, having received an affirmative response, asks further, *Then, should people be allowed to yell "Fire" in a crowded auditorium.*

During the discussion we note that putting these questions in just this way reflects a peculiar affront to the practice of deductive thinking. Subsequently, we ask if anyone can explain the fallacy indicated by this particular line of questioning. With little further provocation the students quickly point out that given the word *freedom* means *absence of restraint*, then, if there is to be an absence of restraint upon a person's speech then a person must, logically speaking, be free to shout *Fire* or any other such thing when and where they choose. Logically speaking one cannot aver that there must be freedom of speech and then proceed to list prohibitions against it.

At this point we find we ask the children what the point is to having a constitutional provision regarding our acts of speech. Without fail the children quickly begin by noting that it would be imprudent to create a license to speak so unbridled that the welfare of others may be directly imperiled as a result. Instead, they conclude that the point behind such a constitutional provision is to create a right which delimits the restraints that might be imposed on a given speaker or a given set of ideas. Thus, a discussion about constitutional protections becomes a lesson in logical analysis and a lesson in logical analysis illustrates the point behind initiating certain constitutional provisions. The children can now address directly the positive concern about which specific speech activities it is important to protect and how should a constitutional provision be interpreted so as to guarantee their protection.

Such discussions are far less animated than those which typically attempt to detail the rationale behind our constitutional protections. Nevertheless, we have found that children are intensely interested in both the process and outcome of such discussions and, more importantly perhaps, they learn the extent to which skills of logical analysis can be employed to make progress in understanding an abstract issue of immediate social import.

There is no area of the gifted student curriculum that cannot be similarly enhanced by supplementing traditional instruction with training in formal logic. For example, in the sciences the

teacher can draw the student's attention to a basic assumption regarding the reoccurrence of a particular phenomenon: There exists a field of attraction between any two bodies. The teacher can then explain that even though every test of a hypothesis takes the form of a deductive (i.e., truth-guaranteeing) argument the **assumptions** which initiate the hypothetical explanation must be the product of some prior conjecture on the part of a scientist. The children are then asked to consider the events that lead a scientist to make the assumptions that he or she makes.

Typically, assumptions such as the law of gravity have been well confirmed by the scientists' experiences. But no scientist can have experiences of what will happen tomorrow (or at any other point in space-time with which he is unfamiliar) so generalized statements such as the law of gravity must be the product of an inductive inference. Subsequent to such discussion, students quickly begin to understand that both induction and deduction are necessary elements at different stages in the practice of experimental science. They recognize that the certainty which seems to accompany scientific conclusions is largely the result of the deductive form of argument characteristic of reports of the consequences of experimental research. Similarly, they recognize that the reason all scientific conclusions must be held tentatively is that the assumptions of science are the product of inductive thinking and as such necessarily contain an element of uncertainty.

Conclusion

Clearly, there is much to be gained in every area of the gifted curriculum by employing the definitions of deductive and inductive inference-making characteristic of the most recent advances in logic. In the face of such evidence it would be inexcusable to continue to fetter the child's understanding of inference-making by employing the Aristotelian definitions of deduction and induction that were so common in this country 50 years ago. As the formal study of logic advances so too must our understanding of its role in pedagogy.

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Training Analogical Reasoning Skills in the Gifted

Patricia A. Alexander

This article describes a training program in analogical reasoning skills based on Sternberg's componential processes research. Students are first taught how to apply the processes of encoding, inferring, mapping and applying in the solution of linguistic and nonlinguistic analogies of the A:B::C:D type. Then, they apply these components in the processing of extended linguistic analogies developed in sentence, paragraph, and story form. Finally, several weeks are spent teaching students the utility of these same components and analogical reasoning to their performance in the content areas.

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While such concerns as identification procedures, grouping alternatives, and resource allocation are vital to the success of any gifted program, its heart remains the interaction between teacher and learner. The ultimate measure of any program should be its ability to deliver effective instruction to those learners it is intended to serve. Effective in-

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struction is an active process that involves *calculated models, suggestions, rules or injunctions that have a known influence on the way a child thinks* relative to some specific task (Belmont & Butterfield, 1977, p. 446). One instructional focus gaining favor in gifted programs is the training of intellectual skills. Following a quiescent period, there is now a burgeoning of programs aimed at enhancing the reasoning abilities of learners (Sternberg, 1983). This article describes one such training program which provides learners with direct instruction in analogical reasoning and does so in a way that satisfies the following criteria for effective instruction:

- **The program should be theoretically sound** (Belmont & Butterfield, 1977; Sternberg, 1983);
- **The program should provide for the explicit translation of empirically-verified processes into experiences that are culturally and developmentally relevant** (Belmont & Butterfield, 1977; Duffy, 1981; Sternberg, 1981);
- **The program should incorporate opportunities for transfer and generalization of training with links to real-world behaviors** (Brown, Campione, & Murphy, 1977; Garner, Wagoner & Smith, in press; Sternberg, 1983);
- **The program should be assessed not only for immediate, direct effects but also for the transferability and durability of training** (Kuhn, 1974; Sternberg, 1983; Garner, Hare, Alexander, Haynes & Winograd, Note 1).

In describing the elements of the training program in analogical reasoning, each of these criteria will be addressed.

The Theoretical Base

There are few intellectual skills as pervasive in or as essential to one's existence as the ability to reason analogically. We reason analogically whenever we determine that some conceptual or perceptual experience parallels another. The term, **analogy**, is applied to a wide range of cognitive activities. Broadly defined, it describes a general learning strategy whereby unfamiliar stimuli are introduced with some reference to the more familiar. Teachers frequently apply this strategy - prefacing a new lesson with a review of known, related knowledge (Hayes & Tierney, 1980; Tierney & Cunningham, 1980; Marr & Gormley, 1982). At its most specific, analogy refers to a particular type of reasoning problem of the form, A:B::C:D.

According to Sternberg (1977, 1980,

1981, 1982), perhaps the most influential researcher of analogies, the ability to reason analogically is predicted upon the learner's ability to perform the component processes of **encoding, inferring, mapping, and applying**. These components represent the fundamental units of processing, responsible for what is called intelligent behavior (Sternberg, 1977, p. 317). Simply translated, these components involve the following actions:

Encoding - The attribute identification component; the representation of identifiable characteristics of each term in the analogy. The term, ball, for example, could be described by such attributes as round or bouncy.

Inferring - The determination of a rule that relates adjacent terms. In the analogy, feather:bird::leaf:tree, the relationship between feather and bird is that feather is part of a bird (feather grows on bird).

Mapping - The discovery of relationships between non-adjacent terms. In the analogy, king:prince::queen:princess, king and queen are both members of royalty differentiated on the basis of gender.

Applying - The completion of the analogy; the utilization of derived rules to generate the predicted response term. In the analogy, bat:ball::hammer:_____, the best answer would be nail since bat is used to hit a ball and a hammer is used to hit a nail.

It is these empirically-tested component processes that will form the basis of this training program in analogical reasoning.

Explicit Instruction

As noted, effective instruction requires the appropriate translation of theory into explicit instructional content. Therefore, while the component processes of analogical reasoning serve as the content of instruction, the principles of direct instruction structure the delivery of that content. According to the principles of direct instruction, the training program is teacher-directed; that is, the presentation and pacing of the lessons are under the instructor's control (Duffy, 1981; Rosenshine, 1979). Toward this end, a three-day initial training phase was prepared in scripted form.* By scripting the initial phase of training, there can be adequate guidance to the teacher, as well as some control for information presentation.

Another principle of direct instruction includes extensive coverage of

*The script of the initial training phase is available upon request from the author.

content, sequenced from simple to complex. For this purpose, Session One of the training moves from instruction in nonlinguistic, concrete analogies to nonlinguistic, abstract analogies, both of the A:B::C:D format. The concrete, nonlinguistic analogy problems are simple drawings of familiar objects (ex., ☐ : ☐ :: ☐ : ☐), while abstract, nonlinguistic problems consist of geometric figures (ex., △ : ▲ :: □ : ■). Within each set, concrete and abstract, there is an attempt to move from easier to more difficult problems.

Those applying this type of training program must take care that the selection of sample problems, and the pacing of material be determined by the abilities, interests, and maturity of the learners involved. While commercial materials may be available for use in the training, especially with older learners, the author has found it advisable to generate practice problems. Analogy problems can be easily developed using available classroom materials (ex., pictures, drawing from unused readers, workbooks). Older students and parents can assist in the development of training materials. Test materials, such as the Lorge-Thorndike Intelligence Tests (Lorge, Thorndike & Hagen, 1964) and the Cognitive Abilities Test (Thorndike & Hagen, 1978), serve as ready sources of model analogy problems.

In **Session One**, instruction begins with an explanation to the students of training goals, and a definition of analogy. Students are told that analogies are a special type of reasoning problem that can be solved by following specific steps. At this point, the component processes are introduced and described. Students then rehearse the naming and explaining of each component. Once students demonstrate understanding of the components, this knowledge is applied, first, to a series of concrete, nonlinguistic analogy problems, and, then, to a series of abstract, nonlinguistic analogy problems. The first session is structured to provide group practice, followed by independent practice.

Another principle of direct instruction specifies that students' progress should be monitored and instruction appropriately adapted. In addition, feedback to the student should be immediate and academically-oriented (Duffy, 1981; Rosenshine, 1979; Samuels, 1981). These principles, as well, have been incorporated in the analogical reasoning training program. During the group practices, students are shown sample analogy problems, term by term, along with the appropriate processing directive, to which they re-

spond verbally. For example, on the directive to encode, students describe the attributes of each term as displayed. On the directive to infer, they specify the relationship between adjacent terms. When directed to map, students predict the relationship of the nonadjacent terms. Finally, on the directive to apply, students furnish the response that best completes the analogy. This type of explicit, oral practice is viewed as essential in the training program to deautomatize and externalize the reasoning process.

Session Two of the initial training period provides direct instruction in the component processes applied to linguistic analogies, again of the A:B::C:D format (ex., sailor:ship::flyer:airplane). Session Two begins with a review of the previous day's activities, including the completion of a practice sheet in nonlinguistic analogies. The employed procedure parallels that applied in the first session, with the inclusion of group and independent practice, evaluation and feedback.

In Session Three of the training, students receive instruction in analogies incorporated in larger text units, beginning with sentences (ex., From his vantage point, the mountains stood tall while the valleys sank low against the sky), then paragraphs and moving into whole stories (ex., How does the movie ET compare to the story the Wizard of Oz?). Throughout Session Three, the role of the component processes in the comprehension of extended analogies is reinforced.

Transfer and Durability of Training

At the outset of training, students are informed that the instruction they will receive has two purposes: 1) Students should perform better on specific analogy problems which are important items in school tests; and, 2) They should be able to use these components in analogical reasoning to perform better in their school subjects, such as reading. **Session Three** of the training specifically focuses on reading and how successful reading requires the same component processes of encoding, inferring, mapping and applying. During the extensive and reinforcement phase of the analogy training (which extends several weeks past the initial training), students are shown the utility of these reasoning components in such content areas as math, social studies, and science. For example, equivalent fractions are described as mathematical analogies, comparison/contrast maps

are drawn for particular historical events, objects, or situations, and in science, characteristics or properties of objects are encoded. Teachers and students are encouraged to apply the processes in analogical reasoning to as many diverse tasks as possible.

Program Evaluation

As noted, appropriate evaluation of this training program, or any other intellectual skills training program, should involve the assessment of the direct and indirect effects of instruction. For the current program, the direct effects of training on analogical reasoning have been examined by assessing students' pre-training and post-training performance on analogy problems of the A:B::C:D type. The data on the transfer effects of training have been gathered from a wide variety of sources, including performance on measures of reading comprehension and nonlinguistic intelligence tests.

Data from the several instances of use (cf. Alexander, White & Mangano, 1982; White & Alexander, in press) have demonstrated statistically significant effects for training on students' performance on simple analogy problems. For example, assessment of the fourth-grade population currently undergoing the extension and reinforcement phase of the training, shows a significant gain on the Analogy Test A and B, based on the word comprehension subtest of the Woodcock Reading Mastery Tests, Forms A and B (Woodcock, 1973), $F(1,59) = 63.36, p < .001$. These learners' average number of correct responses on the 30 item test went from 5.16 on the pretest to 13.76 on the posttest (students were tested one week after the initial training period). Preliminary data from the extension and reinforcement phase of training nearing completion are also encouraging. As the application of the analogical reasoning program to learners of varying abilities and ages is continued, more data on its direct and indirect effects will be amassed.

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